

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination

University Theory Examination Nov/Dec 2017

PH204 Pharmaceutical Analysis I

Date: 01/12/2017, Friday

Time: 10:00am to 1:00pm

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

A Give a brief note on:

- i) Common ion effect 05
- ii) Ionic product of water 05

B Define buffer. Derive Henderson-Hasselbach equation. 10

- C
- i. What is error? Explain different types of errors. 07
 - ii. Enlist analytical method validation parameters 03

Q 2 Attempt any TWO of the following

A

- i. Define a) Reduction, b) Redox Potential, c) Oxidation Number 04
- d) Self Indicator

ii. Write a note on Diazotization nitrite titration 06

B Differentiate Iodometry and Iodimetry. Discuss iodometric titration with example. 10

C What is complexometric titration? Give the different types of 10
complexometric titration and discuss replacement titration with example.

SECTION- II

Q 3 Attempt any TWO of the following

- A
- i. State the principle underlying Non-Aqueous titration. 03
 - ii. Explain the Differentiating and leveling effects of solvents in non-aqueous titration with suitable example. 07

B Explain Mohr's method and Fajan's method of precipitation titration. 10

C Explain the steps involved in gravimetric analysis. 10

Q 4 Attmpt any TWO of the following

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|---|------|--|----|
| A | i. | State the difference between solid liquid extraction and liquid liquid extraction. | 02 |
| | ii. | Explain the process of continuos extraction . | 08 |
| B | i. | State the principle of gravimetry. | 02 |
| | ii. | Explain the types of precipitation in gravimetry | 05 |
| | iii. | Explain factors affecting solubility. | 03 |
| C | | Explain the principle of Karl Fisher titration with chemical reactions. | 10 |
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